Exp. 4(BCS551)-DQL and Sorting Data

Objective:At the end of the assignments, participants will be able to understand basics of DQL commands, Select, Conditional retrieval, operators, pattern matching, order by clause.

Using the where clause:

Query1:

Select employee_id,last_name,job_id,department_id
From hr.employees
where department_id=90;

Output1:

Download ▾ Execution time: 0.006 seconds

	EMPLOYEE_ID	LAST_NAME	JOB_ID	DEPARTMENT_ID
1	100	King	AD_PRES	90
2	101	Yang	AD_VP	90
3	102	Garcia	AD_VP	90

Query2:

Select last_name,job_id,department_id
From hr.employees
where last_name='Whalen';

Output2:

Download ▾ Execution time: 0.006 seconds

	LAST_NAME	JOB_ID	DEPARTMENT_ID
1	Whalen	AD_ASST	10

Query3:

Select last_name,salary
From hr.employees
where salary<=3000;

Output3:

Download Execution time: 0.005 seconds

	LAST_NAME	SALARY
1	Baida	2900
2	Tobias	2800
3	Himuro	2600
4	Colmenares	2500
5	Mikkilineni	2700
6	Landry	2400
7	Markle	2200
8	Atkinson	2800
9	Marlow	2500
10	Olson	2100

Query4:

Select last_name,salary
From hr.employees
where salary between 2500 and 3500;

Output4:

Download Execution time: 0.005 seconds

	EMPLOYEE_ID	LAST_NAME	SALARY	MANAGER_ID
1	101	Yang	17000	100
2	102	Garcia	17000	100
3	114	Li	11000	100
4	120	Weiss	8000	100
5	121	Fripp	8200	100
6	122	Kaufling	7900	100
7	123	Vollman	6500	100
8	124	Mourgos	5800	100
9	145	Singh	14000	100
10	146	Partners	13500	100

Query5:

Select employee_id,last_name,salary,manager_id
From hr.employees
where manager_id in (100, 101, 201);

Output5:

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	LAST_NAME	SALARY
1	Khoo	3100
2	Baida	2900
3	Tobias	2800
4	Himuro	2600
5	Colmenares	2500
6	Nayer	3200
7	Mikkilineni	2700
8	Bissot	3300
9	Atkinson	2800
10	Marlow	2500

Query6:

Select first_name
From hr.employees
where first_name like ’S%’;

Output6:

Download Execution time: 0.006 seconds

	FIRST_NAME
1	Sundar
2	Shelli
3	Sarah
4	Shelley
5	Susan
6	Steven
7	Sundita
8	Steven
9	Samuel
10	Sarath

Query7:

Select last_name
From hr.employees
where last_name like '_o%';

Output7:

Download Execution time: 0.005 seconds

	LAST_NAME
1	Colmenares
2	Doran
3	Fox
4	Johnson
5	Jones
6	Mourgos
7	Popp
8	Rogers
9	Tobias
10	Vollman

Query8:

Select last_name,manager_id
From hr.employees
where manager_id is null;

Output8:

Download Execution time: 0.004 seconds

	LAST_NAME	MANAGER_ID
1	King	*(null)*

Using the ORDER BY Clause

Query9:

```
Select last_name, job_id, department_id, hire_date
FROM hr.employees
ORDER BY hire_date
```

Output9:

Download ▾ Execution time: 0.008 seconds

	LAST_NAME	JOB_ID	DEPARTMENT_ID	HIRE_DATE
1	Garcia	AD_VP	90	1/13/2011, 12:00:0
2	Gietz	AC_ACCOUNT	110	6/7/2012, 12:00:00
3	Brown	PR_REP	70	6/7/2012, 12:00:00
4	Jacobs	HR_REP	40	6/7/2012, 12:00:00
5	Higgins	AC_MGR	110	6/7/2012, 12:00:00
6	Faviet	FI_ACCOUNT	100	8/16/2012, 12:00:0
7	Gruenberg	FI_MGR	100	8/17/2012, 12:00:0
8	Li	PU_MAN	30	12/7/2012, 12:00:0

Query10(Sorting in descending order):

```
Select last_name, job_id, department_id, hire_date
FROM hr.employees
ORDER BY hire_date DESC;
```

Output10:

Download ▾ Execution time: 0.005 seconds

	LAST_NAME	JOB_ID	DEPARTMENT_ID	HIRE_DATE
1	Kumar	SA_REP	80	4/21/2018, 12:00:0
2	Banda	SA_REP	80	4/21/2018, 12:00:0
3	Ande	SA_REP	80	3/24/2018, 12:00:0
4	Markle	ST_CLERK	50	3/8/2018, 12:00:00
5	Lee	SA_REP	80	2/23/2018, 12:00:0
6	Philtanker	ST_CLERK	50	2/6/2018, 12:00:00
7	Geoni	SH_CLERK	50	2/3/2018, 12:00:00
8	Zlotkey	SA_MAN	80	1/29/2018, 12:00:0
9	Marvins	SA_REP	80	1/24/2018, 12:00:0

Query11:

Select employee_id,last_name, salary*12 annsal
FROM hr.employees
ORDER BY annsal;

Output11:

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	EMPLOYEE_ID	LAST_NAME	ANNSAL
1	132	Olson	25200
2	128	Markle	26400
3	136	Philtanker	26400
4	135	Gee	28800
5	127	Landry	28800
6	119	Colmenares	30000
7	131	Marlow	30000
8	140	Patel	30000
9	144	Vargas	30000

Query12:

Select last_name,department_id, salary
FROM hr.employees
ORDER BY department_id,salary DESC;

Output12

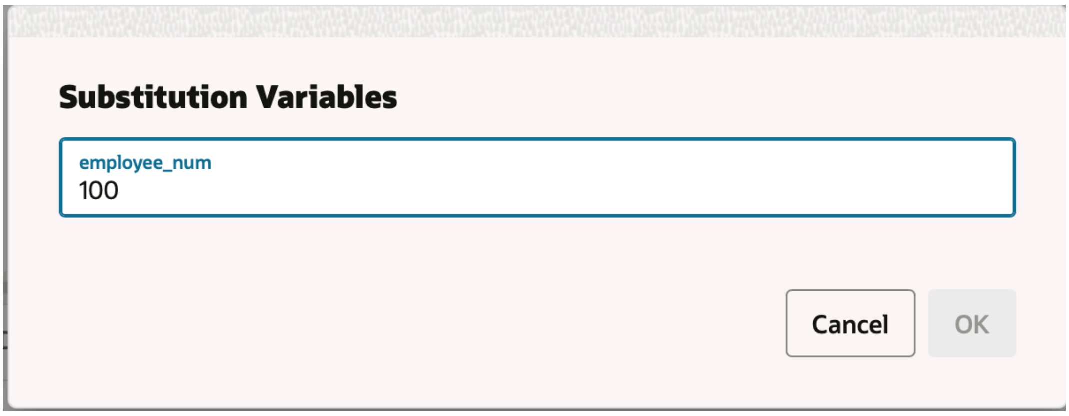
Download ▾ Execution time: 0.007 seconds

	LAST_NAME	DEPARTMENT_ID	SALARY
1	Whalen	10	4400
2	Martinez	20	13000
3	Davis	20	6000
4	Li	30	11000
5	Khoo	30	3100
6	Baida	30	2900
7	Tobias	30	2800
8	Himuro	30	2600
9	Colmenares	30	2500

Query13:

```
Select employee_id,last_name, salary, department_id
FROM hr.employees
where employee_id=&employee_num
```

Output13:



	EMPLOYEE_ID	LAST_NAME	SALARY	DEPARTMENT_ID
1	100	King	24000	90

Query14:

```
SELECT last_name, department_id, salary * 12
FROM hr.employees
WHERE job_id = ‘&job_title’;
```

Output14:

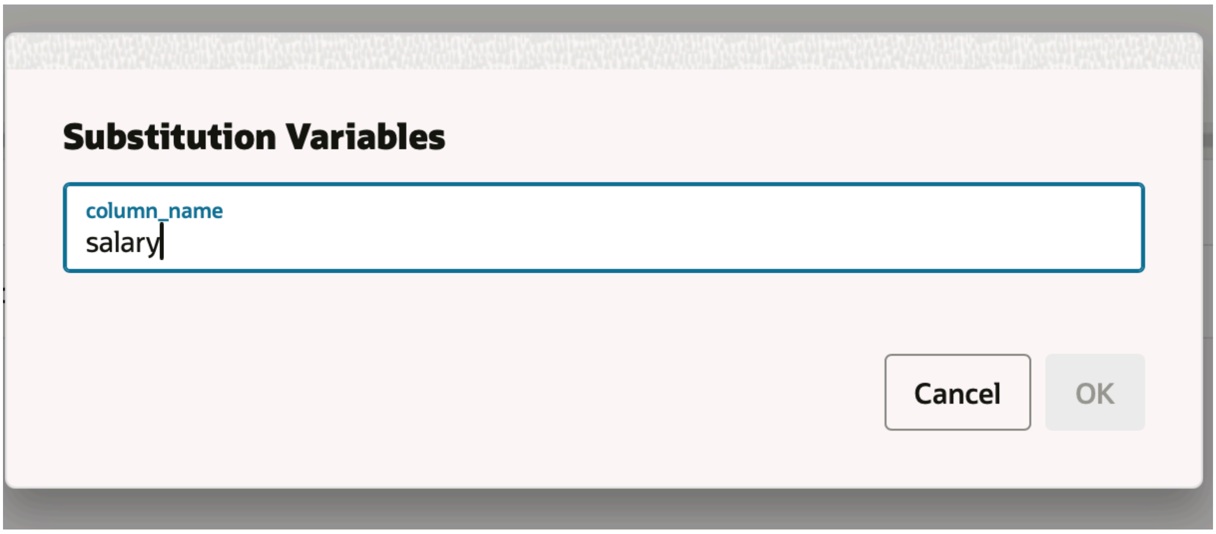
Download ▼ Execution time: 0.005 seconds

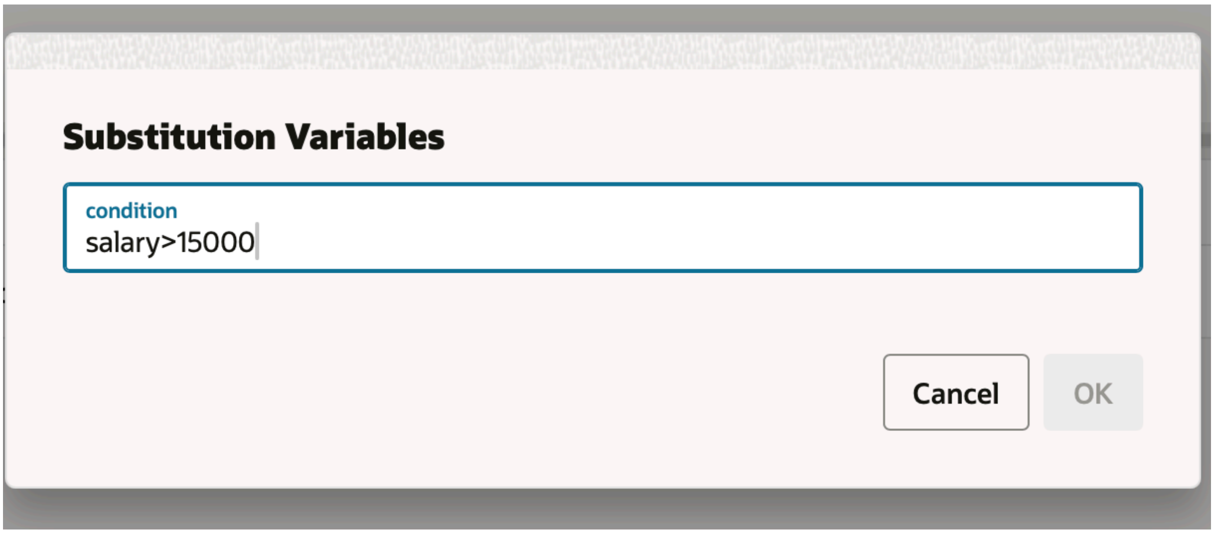
	LAST_NAME	DEPARTMENT_ID	SALARY*12
1	James	60	108000
2	Miller	60	72000
3	Williams	60	57600
4	Jackson	60	57600
5	Nguyen	60	50400

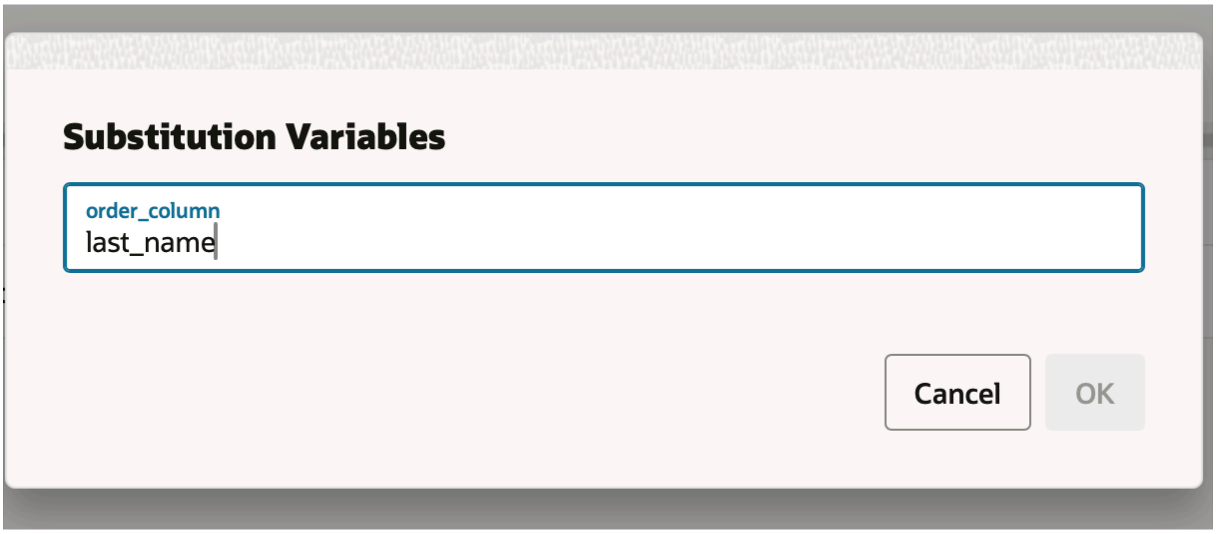
Query15:

Select employee_id,last_name, job_id, &column_name
FROM hr.employees
WHERE &condition
Order by &order_column;

Output15:







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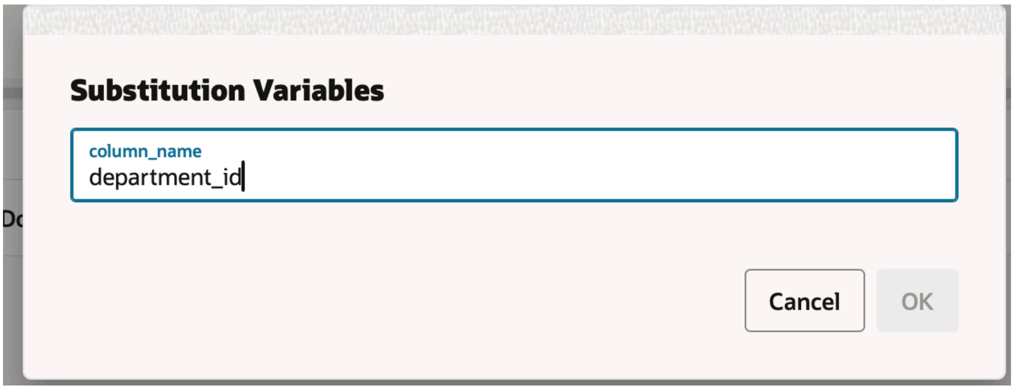
	EMPLOYEE_ID	LAST_NAME	JOB_ID	SALARY
1	102	Garcia	AD_VP	17000
2	100	King	AD_PRES	24000
3	101	Yang	AD_VP	17000

Query16:

Select employee_id,last_name, job_id, &&column_name
FROM hr.employees
Order by &column_name;

Output16:



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	EMPLOYEE_ID	LAST_NAME	JOB_ID	DEPARTMENT_ID
1	200	Whalen	AD_ASST	10
2	201	Martinez	MK_MAN	20
3	202	Davis	MK_REP	20
4	114	Li	PU_MAN	30
5	115	Khoo	PU_CLERK	30
6	116	Baida	PU_CLERK	30
7	117	Tobias	PU_CLERK	30
8	118	Himuro	PU_CLERK	30
9	119	Colmenares	PU_CLERK	30
10	203	Jacobs	HR_REP	40